

Video RAG Retrieval Project

<https://github.com/aiyshwary/Video-RAG-Retrieval>

OVERVIEW

A semantic video search system that extracts frames, generates scene captions using ViT-GPT2, and embeds them using SentenceTransformers and OpenCLIP into a dual FAISS index, enabling natural language queries to retrieve relevant video scenes with timestamps.

IMPACT

Built a Visual RAG system that makes unstructured video content semantically searchable: a text query retrieves relevant video timestamps by matching against dual text and image embedding indexes via cosine similarity.

TECHNICAL WALKTHROUGH

Visual RAG Retrieval is a semantic video search system that extracts frames, generates scene captions using ViT-GPT2, and embeds them using SentenceTransformers and OpenCLIP into a dual FAISS index. This enables natural language queries to retrieve relevant video scenes with timestamps.

Uses vision-language models (e.g., Video-LLaMA, ViT-GPT2) to generate natural language descriptions of video scenes.

Extracts key frames from videos at a configurable FPS, storing frame number, timestamp, and associated scene description.

Computes text embeddings (SentenceTransformers MiniLM) for scene descriptions and image embeddings using OpenCLIP for frame images.

Stores embeddings in two parallel FAISS indexes (text and image), each linked by video metadata (video ID and timestamp).

User text queries are embedded and matched against scene and frame embeddings using cosine similarity to retrieve relevant scenes and their timestamps.

i) Dual-modality retrieval: supports both text and image-based queries.

ii) Efficient vector search using FAISS IndexFlatIP with L2-normalized vectors.

iii) Frame sampling rate is configurable to balance redundancy and coverage.

iv) Modular architecture: preprocessing, embedding, storage, and retrieval are decoupled for scalability.

v) Includes a pytest suite and a self-contained demo that synthesizes a 30-frame test video with OpenCV.

i) Find all scenes with a 'red car' (image embedding).

ii) Locate 'a tense conversation' (text embedding).

The system can be extended to use production vector databases (VDMS, Milvus, Pinecone) and advanced search engines (Elasticsearch, Solr).

Visual RAG Retrieval makes video content searchable, explainable, and scalable for applications in surveillance, enterprise video search, and content moderation.

KEY DETAILS

1. Extracts frames at a configurable FPS and generates natural-language scene captions using ViT-GPT4.

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2. Computes text embeddings with SentenceTransformers (MiniLM) and image embeddings with OpenCLIP.
 3. Stores embeddings and metadata in two parallel FAISS indexes supporting independent text or image search.
 4. Query interface accepts a natural language string and returns top-k matching scenes with frame references.
 5. Modular architecture (preprocessing, embeddings, storage, retrieval) designed for extension to production.
 6. FAISS indexes use IndexFlatIP with L2-normalized vectors for cosine-similarity search; OpenCLIP uses ViT-L/14 and LAION-2B with automatic GPU/CPU selection.
 7. Includes a pytest suite and a self-contained demo that synthesizes a 30-frame test video with OpenCV.